

VPBANK TECHNOLOGY HACKATHON 2025

Serverless on AWS



Thanh Dang

Solutions Architect, AWS Vietnam



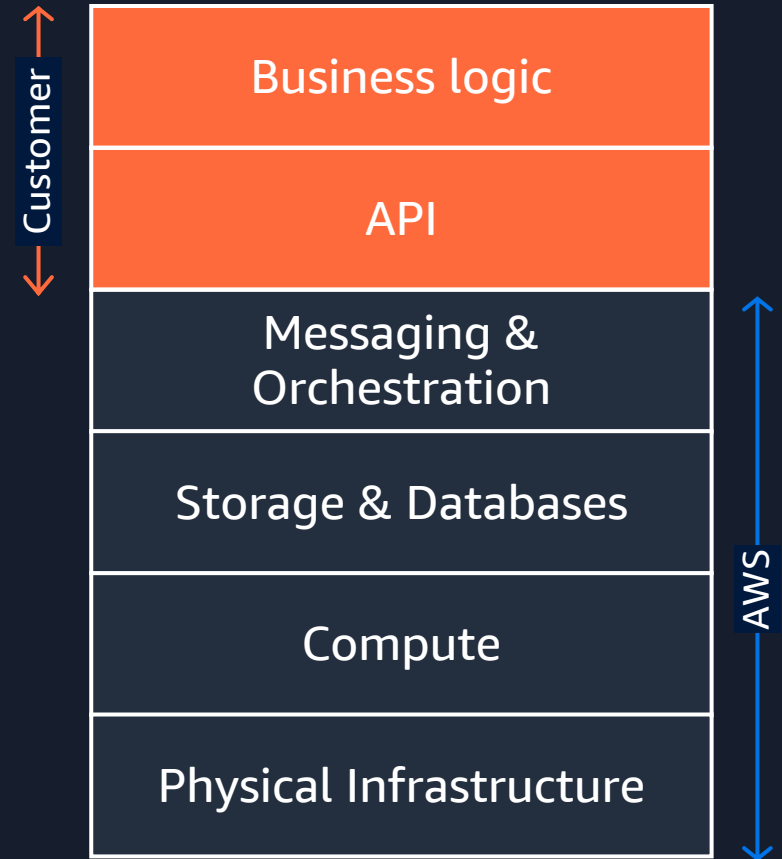
© 2024, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Agenda

- Overview of Serverless on AWS
- Event-Driven Architectures
- Serverless use cases

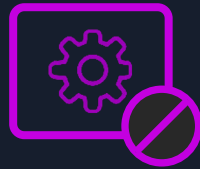
What is Serverless?

Serverless services simplify the management and scaling of cloud applications by shifting undifferentiated operational tasks to the cloud provider so **development teams can focus on writing code** that solve business problems



Why serverless

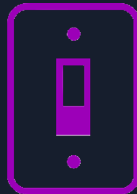
TAKE FULL ADVANTAGE OF THE CLOUD TO MODERNIZE APPLICATIONS AND ACCELERATE INNOVATION



No infrastructure provisioning,
no management



Automatic scaling



Pay for value

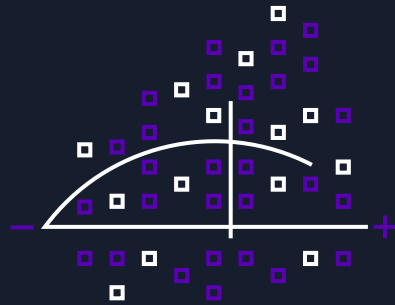


Highly available and secure

What are customers building with Serverless?



**IT
Automation**



**Data
processing**



**Web
applications**



**Machine
Learning**

AWS serverless spectrum

AWS OFFERS THE WIDEST PORTFOLIO OF SERVERLESS SERVICES FOR RUNNING AND BUILDING MODERN APPS

From Primitives to...

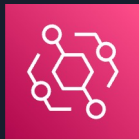
(Examples: compute, containers, buses)



AWS
Lambda



AWS
Fargate



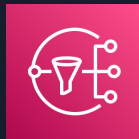
Amazon
EventBridge



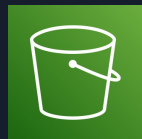
AWS
App
Runner



Amazon
ECS



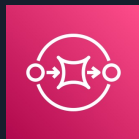
Amazon
SNS



Amazon
S3



Amazon
EFS



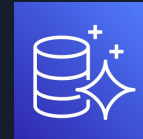
Amazon
SQS



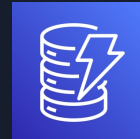
Amazon
API
Gateway

...Peripherals

(Examples: databases, analytics, workflows)



Amazon
Aurora
Serverless



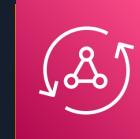
Amazon
DynamoDB



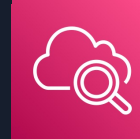
Amazon
Kinesis



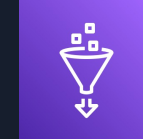
AWS
Step
Functions



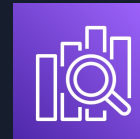
AWS
AppSync



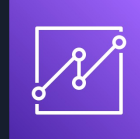
Amazon
CloudWatch



AWS
Glue

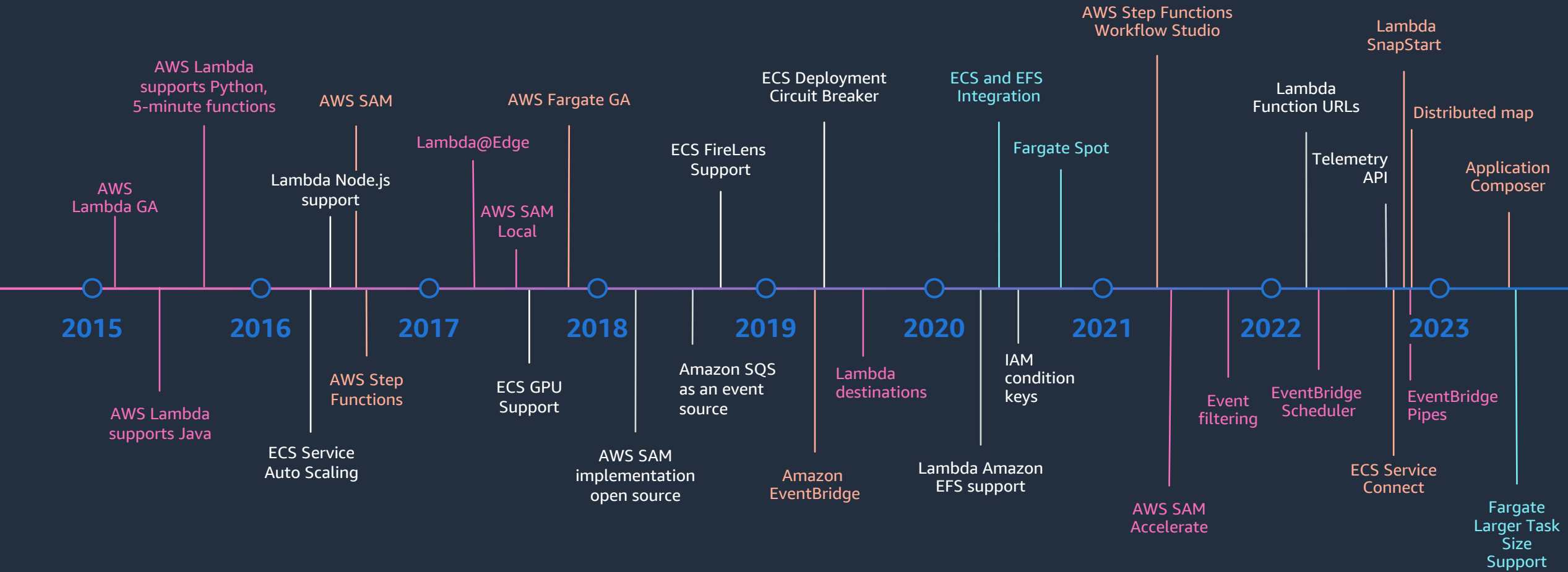


Amazon
OpenSearch



Amazon
QuickSight

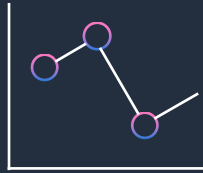
AWS pioneered Serverless and has innovated since Day 1...



Serverless has found broad customer acceptance



Today, **over a million** active customers use Lambda each month

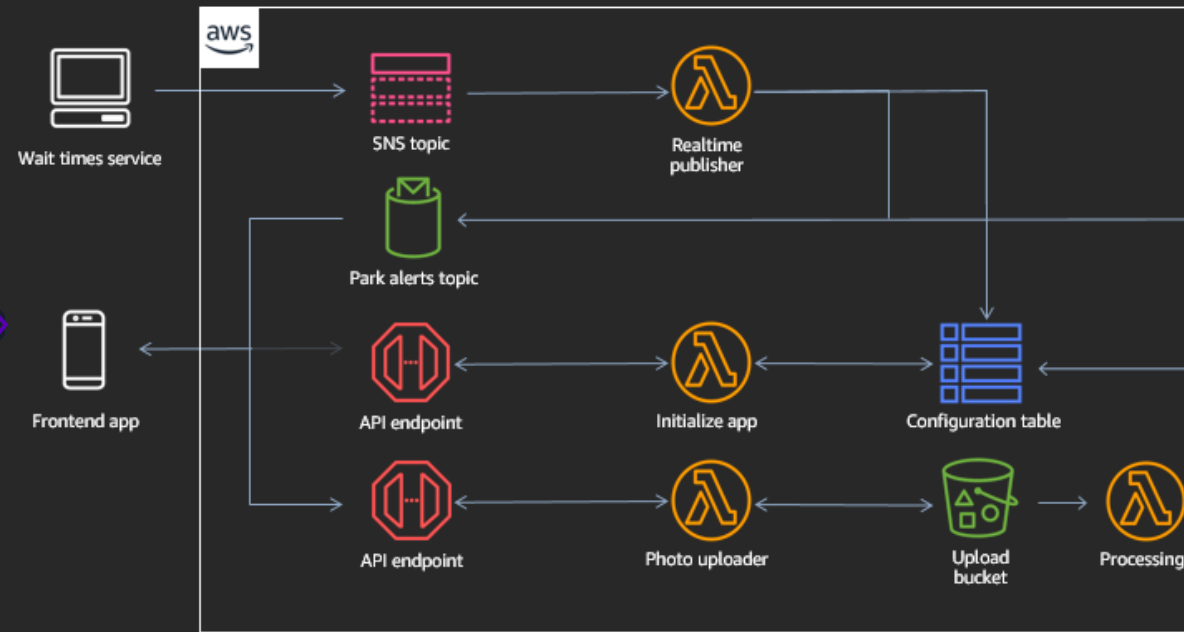
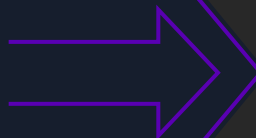


Average number of invokes on EventBridge is over **2 trillion per month**



Over 2.25 billion Amazon ECS tasks are launched each week

Small pieces, loosely joined



AWS Lambda

Event-driven function-as-a-service



Serverless Architecture

Event Source



Changes in data state



Requests to endpoints



Changes in resource state

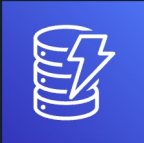
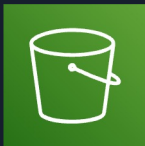


Function



Node.js
Python
Java
C#
Go
Ruby
Bring Your Own

Services / Other



Anatomy of a Lambda Function

Handler function

- Function executed on invocation
- Processes incoming event

Event

- Invocation data sent to function
- Shape differs by event source

Context

- Additional information from Lambda service
- Examples: request ID, time remaining

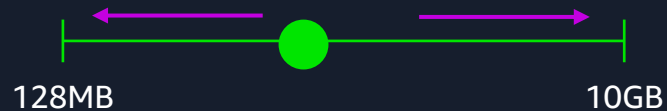
app.py

```
def handler(event, context):  
    msg = 'Hello {}'.format(  
        event['name']  
    )  
    return { 'message': msg }
```

Lambda Function Configuration

Power rating

- Select between 128MB and 10GB
- CPU and network allocated proportionally
- Power tune to balance cost and speed



Permissions model

- **Execution role** grants function access to resources via IAM
- **Resource-based** policy controls invocation from other AWS accounts or organizations and AWS services

Lambda Function Configuration

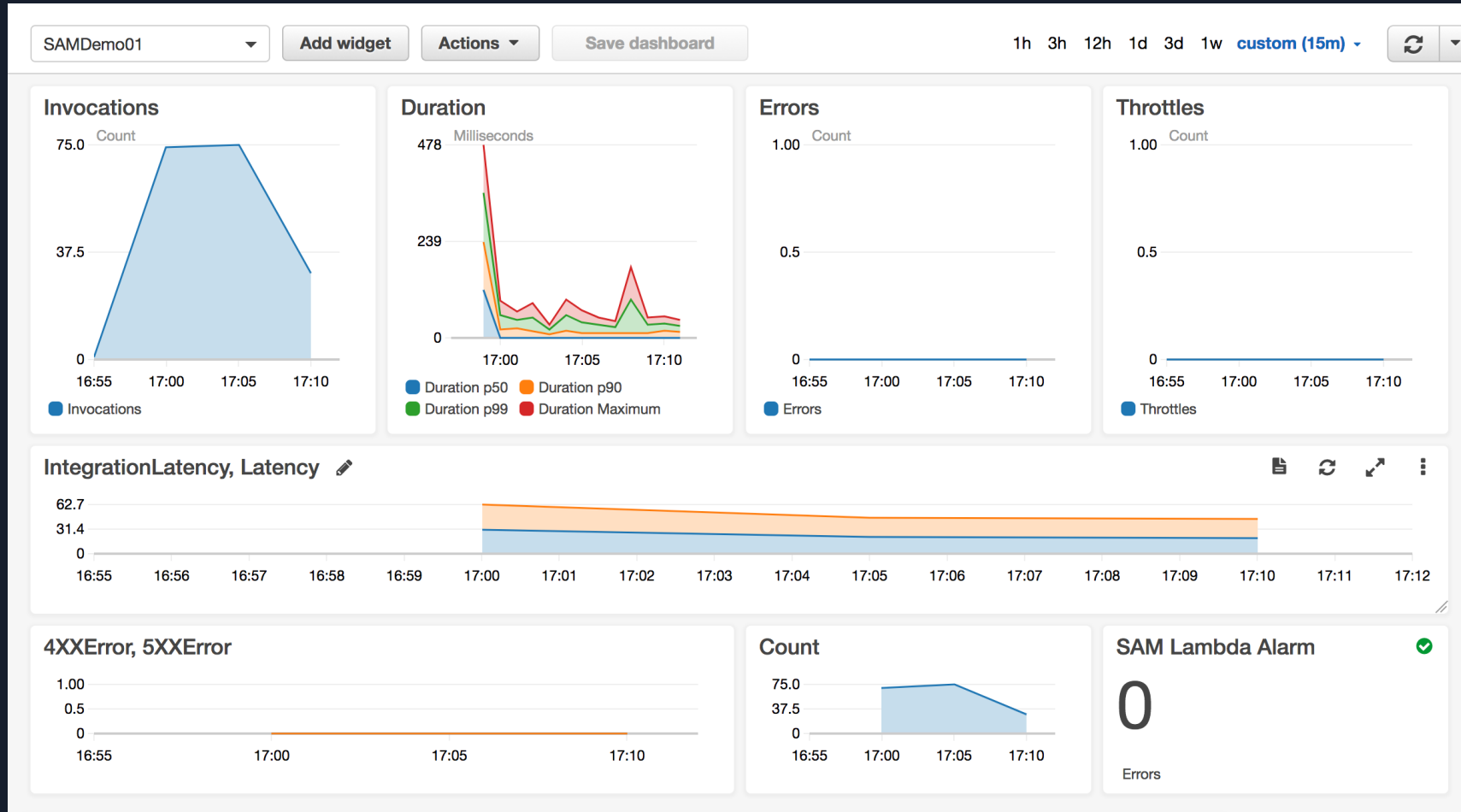
Timeout

- Up to 15 minutes
- Synchronous vs Asynchronous
 - API Gateway timeout = 30 sec

Network access

- Configure access to VPC
- Security Group rules apply
- VPC does **not** enhance security of function

Built in monitoring



Serverless architecture patterns





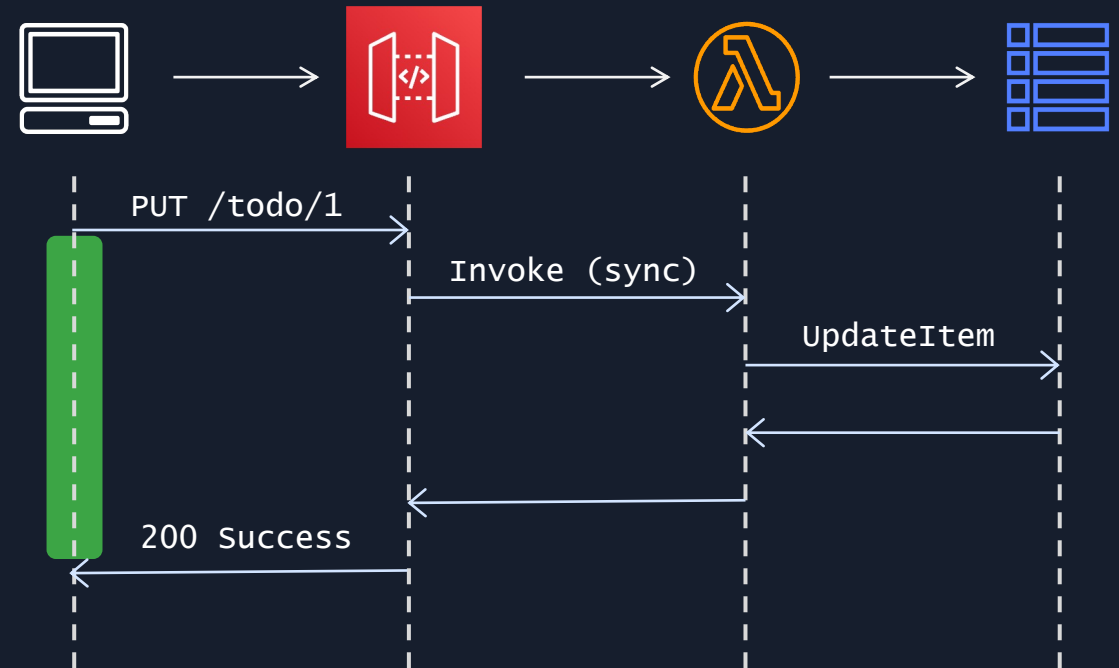
event

[i-'vent] **noun**

A thing that happens,
particularly one of
importance.

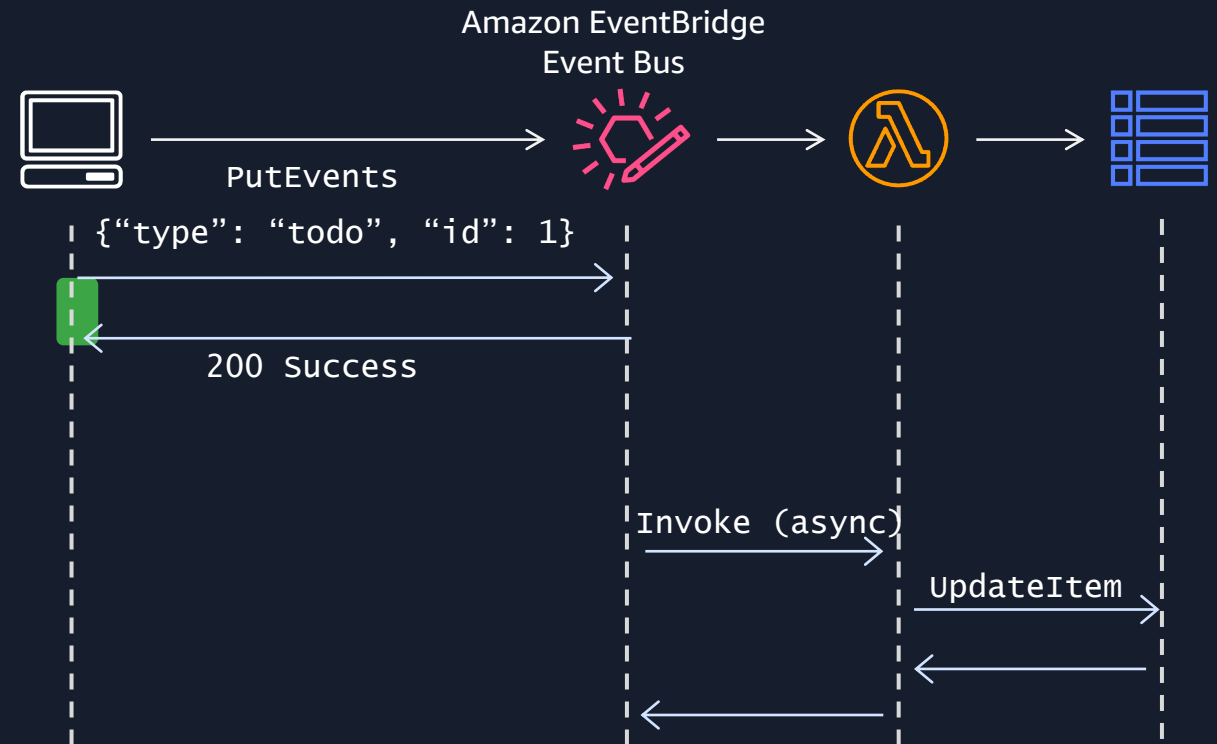
API-Driven Architectures

- API defines the interface
 - e.g. REST, GraphQL
- Caller expects an immediate response
 - Response contains the result of the work
 - Generally, under 30 seconds
 - Synchronous processing
- Client must implement error handling, retry logic



Event-Driven Architectures (EDA)

- Event payload defines the interface
 - e.g. JSON
- Response is “message received” (or not)
 - Asynchronous processing
 - Process can be long running (beyond 30s)
 - Updating client (e.g. UI) can be more challenging
- Higher resiliency, durability
 - Driven by messaging service
- Built-in retries, configurable to use case





events

in **serverless architectures**

Connective tissue of loosely coupled systems.

Represents change in state.
Improving resilience.

Trends of event-driven architectures

WHY CUSTOMERS ARE MOVING TO EVENT-DRIVEN APPLICATIONS

1

Speed & agility

Move faster. Build and deploy services independently.

2

Resiliency

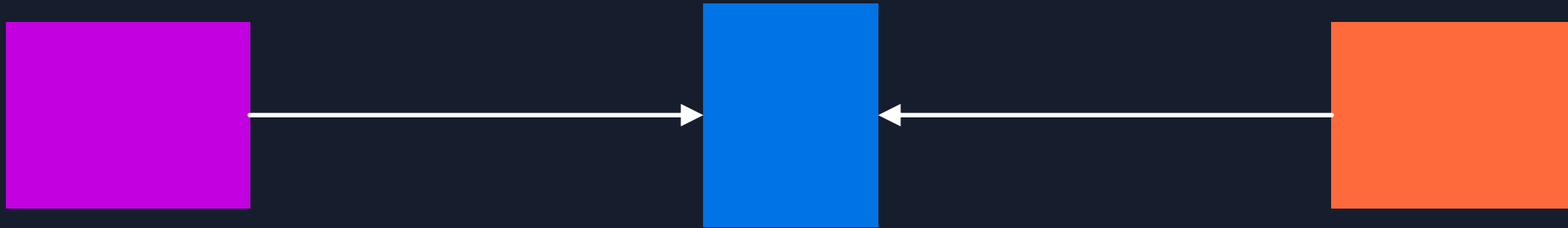
Loosely coupled systems can run and fail independently.

3

Scalability

Minimize waiting time through async and parallel processing.

EDA components



Producer

Event producers are systems that detect a change in state or notice updates and publish those facts. Application code, a database or a time-based trigger can serve as event producers, among other things.

Event broker

An event broker is a meeting place between the producers and consumers. The broker is how events are exchanged.

Consumer

Event consumers are systems that listen for events and use them for their purposes. It's possible, and common, for a consumer to receive an event, perform some work and publish its event in response.

Event brokers

Pull-based

Queues

- Amazon SQS
- Amazon MQ

Streams

- Kinesis
- Kafka/MSK

Push-based

Routers

- EventBridge
- SNS

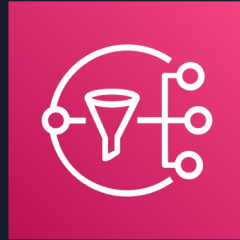
EDA on AWS



Amazon SQS

Messaging

Durable and scalable
Fully managed
Comprehensive security



Amazon SNS

Notifications

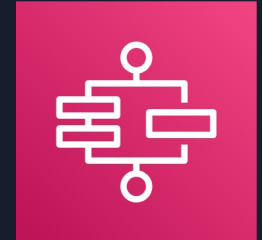
Performance at scale
Fully managed
Enterprise-ready



Amazon
EventBridge

Choreography

Event filtering
Managed & scalable
SaaS integration



AWS
Step Functions

Orchestration

Sequencing
Parallel execution
State management

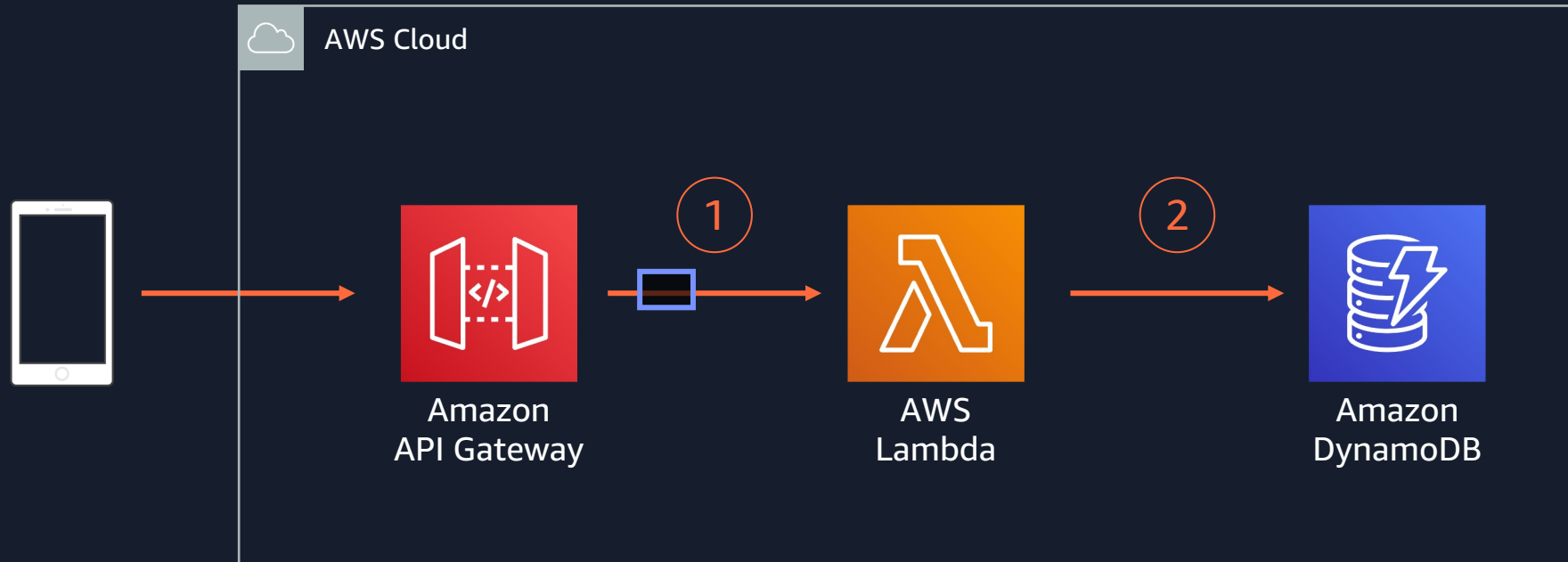
API-Driven use cases

Synchronous processing



RESTful Microservices

HIGHLY-SCALABLE MICROSERVICES

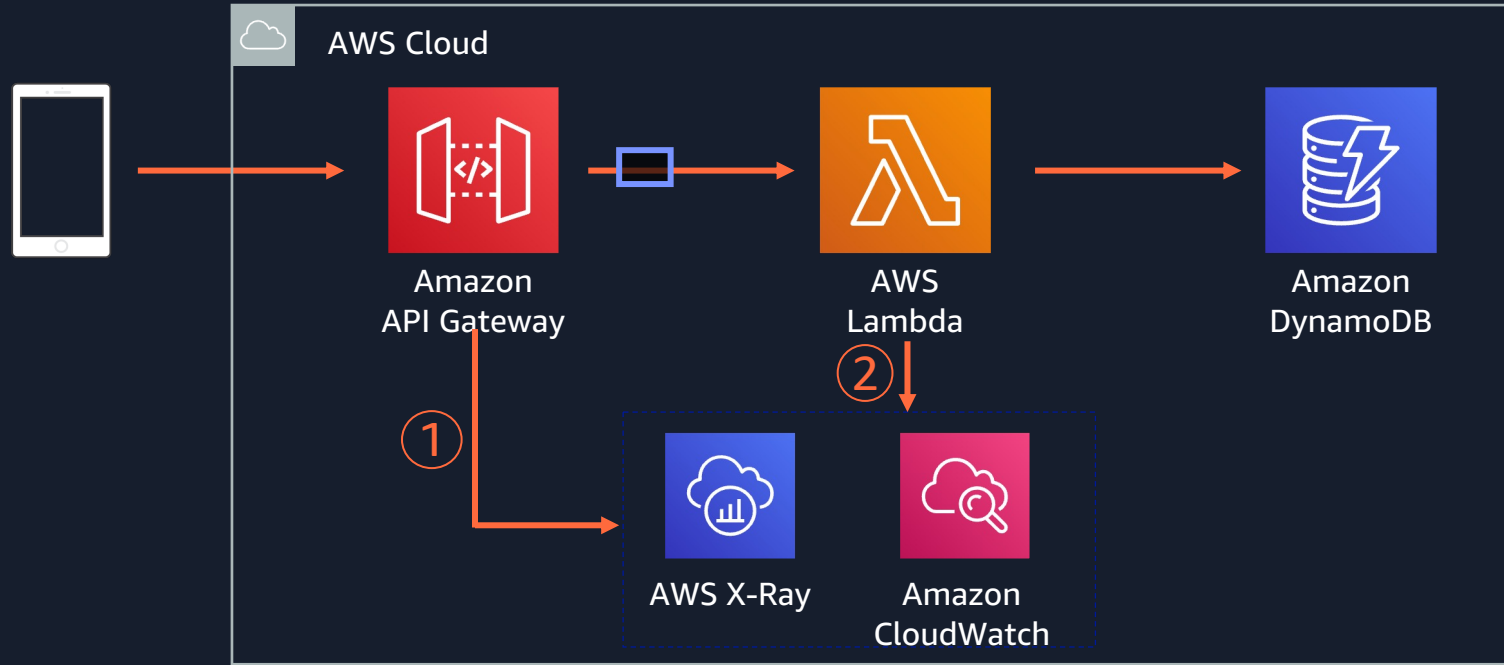


1. API Gateway “translates” incoming HTTP request to event payload

2. Lambda reads / writes data from data store

RESTful Microservices with enhanced observability

ENABLE ACCESS LOGS, STRUCTURED LOGGING, AND INSTRUMENT CODE

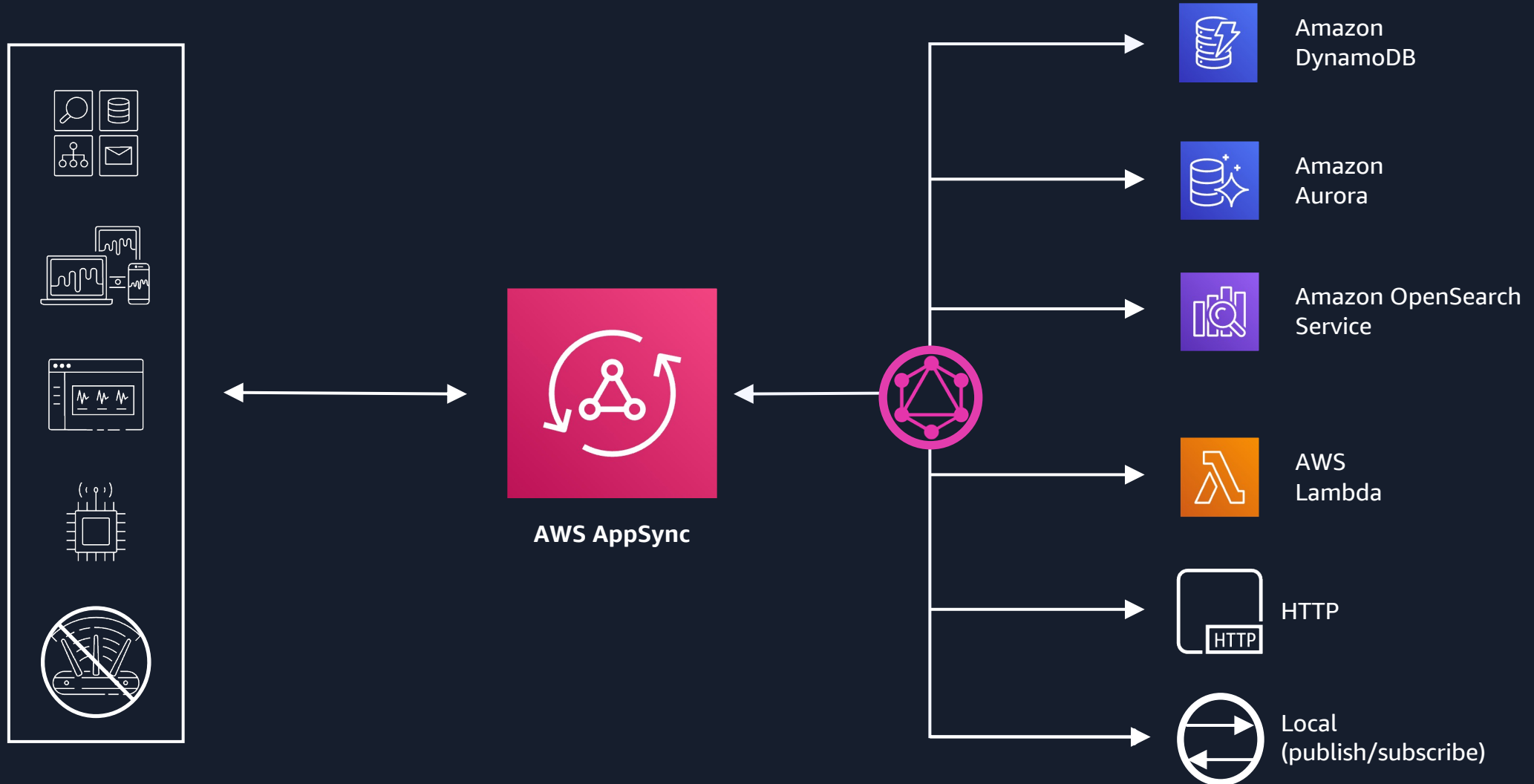


1. Enable access logs and tracing

2. Instrument code and create metrics asynchronously with CloudWatch Embedded Metric Format

AWS AppSync

MANAGED GRAPHQL API FOR UNIFIED DATA ACCESS AND REAL-TIME APPS



Event-driven use cases

Asynchronous processing



Processing file uploads

RESIZE PHOTO, EXTRACT TEXT, TRANSLATE, ETC.



1. Object uploaded to Amazon S3 Bucket
2. **Asynchronous** invoke of Lambda function, event payload includes:
 - Bucket name
 - Object key

Processing file uploads, quickly add new functionality

ADD IMAGE ANALYSIS AND METADATA STORAGE

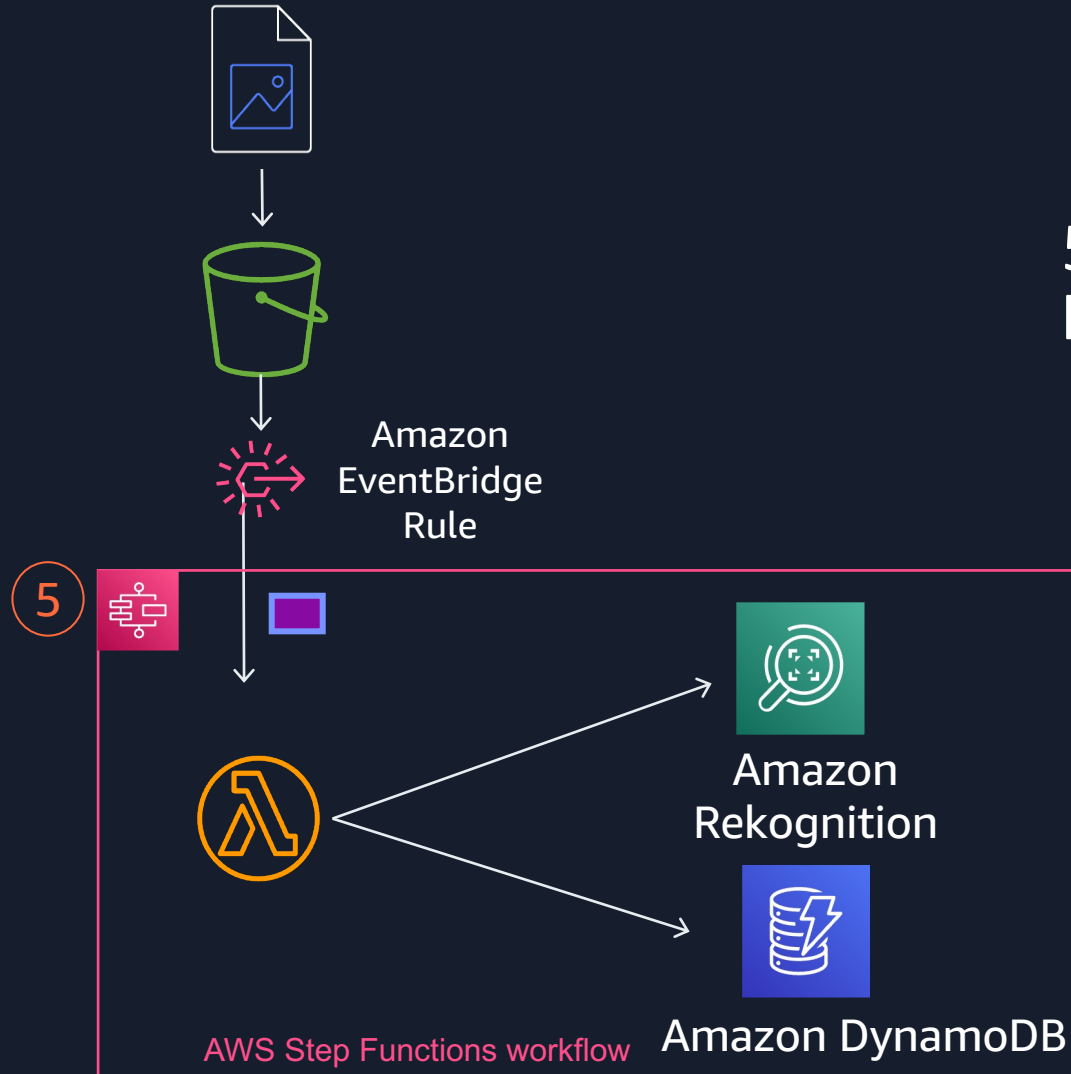


3. Analyze photo with Amazon Rekognition

4. Store image details and results of analysis

Processing file uploads, quickly add new functionality

ADD IMAGE ANALYSIS AND METADATA STORAGE



5. Simplify business logic with Step Functions

Streaming data ingestion and storage

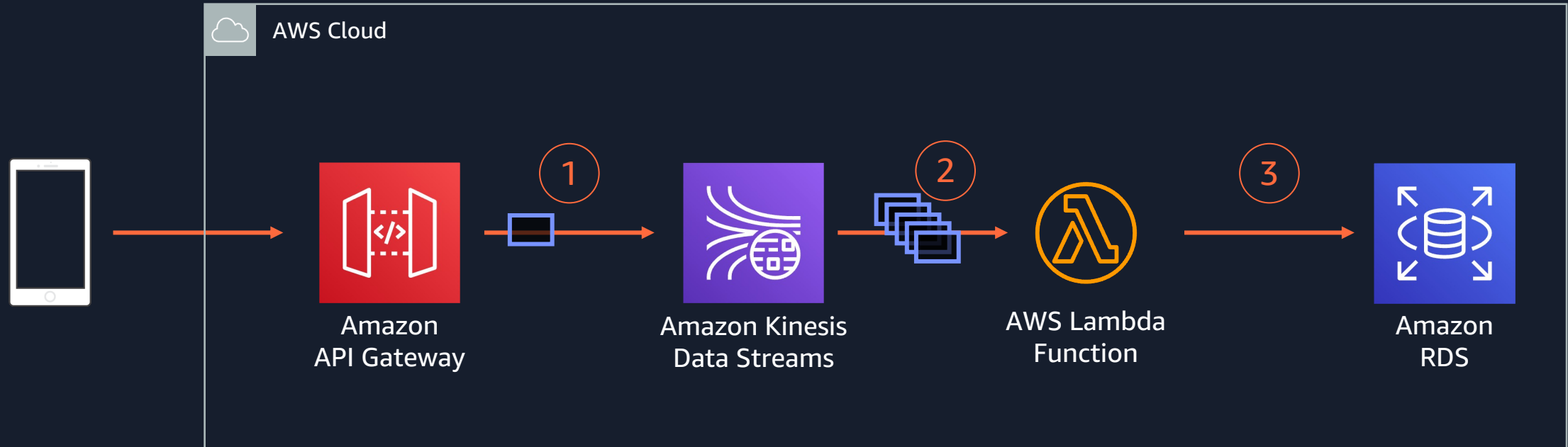
CONSUME, PROCESS, AND STORE DATA



1. Lambda service polls Kinesis Data Stream for messages
2. Function is **synchronously** invoked with **batches** of messages
3. Function processes and/or pushes data to downstream data stores

"Semi-Serverless" Webhook

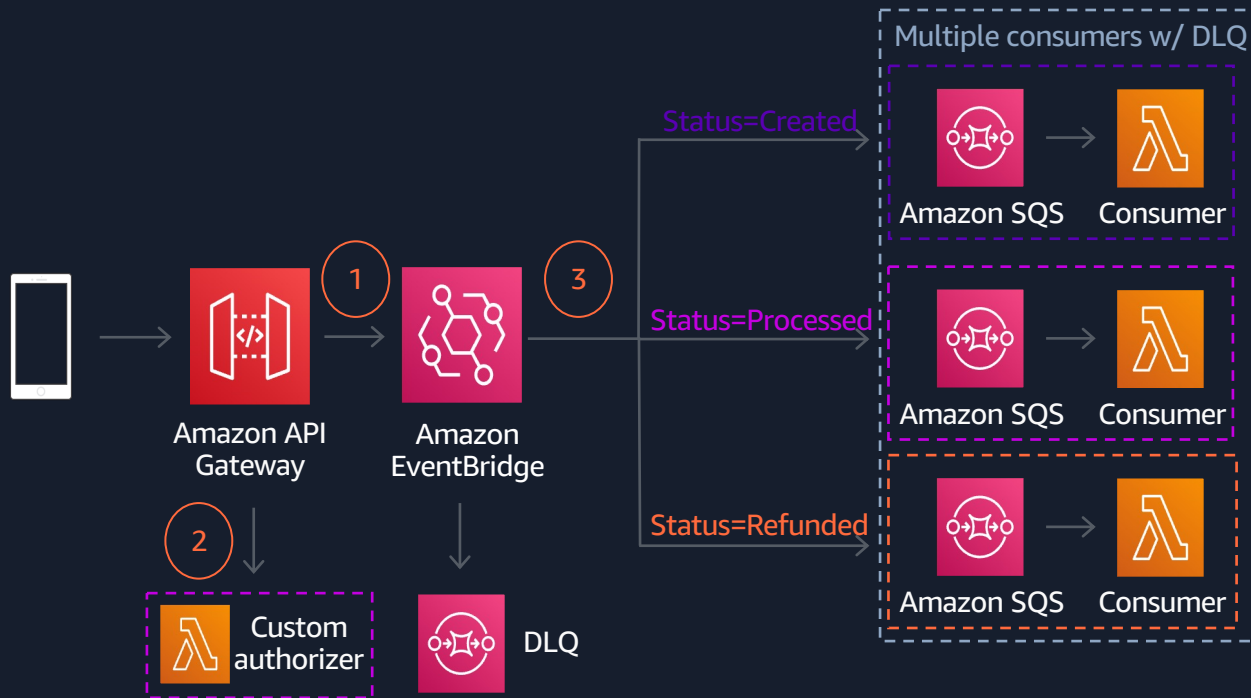
SCALABLE, RESILIENT. BUFFER DOWNSTREAM CONCURRENCY.



1. "Storage-first" pattern: API Gateway writes directly to Kinesis Data Stream
2. Kinesis as a buffer to limit concurrency downstream
3. Perform transaction(s) on batch

Fan out

PUSH UPDATES TO MULTIPLE SUBSCRIBERS

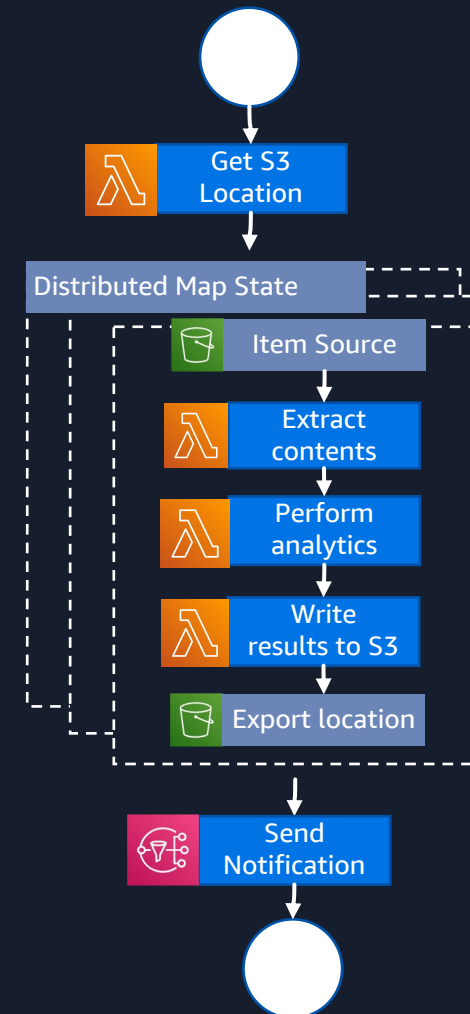


1. “Storage first”: integrate API Gateway directly to EventBridge
2. Enforce authorization
3. Use routing for efficient processing

Large scale data processing with AWS Step Functions

DYNAMICALLY PROCESS LARGE ARRAYS OF DATA WITH DISTRIBUTED MAP STATE

- Coordinate **large-scale parallel workloads** in Step Functions
- Iterate over **millions of S3 objects** such as JSON or CSV files
- Up to **10,000 parallel executions**
- Invoke Lambda or ECS/Fargate for large scale on demand serverless compute



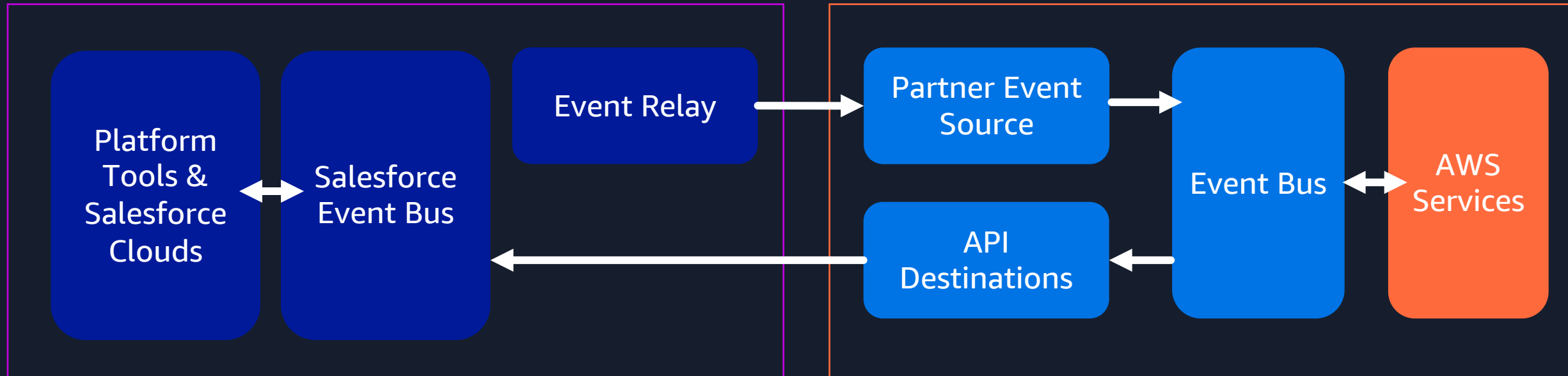
Integrate third party applications with AWS

EXAMPLE: INTEGRATE SALESFORCE WITH AWS

Designed for event driven architecture and real-time applications.

Salesforce

AWS



Continue your AWS serverless learning

Learn at your
own pace



Expand your serverless
skills with our Learning Plan
on **AWS Skill Builder**

Increase your
knowledge



Use our **Ramp-Up Guides**
to build your Serverless
knowledge

Earn AWS
serverless badge



Demonstrate your
knowledge by achieving
digital badges



<https://s12d.com/serverless-learning>

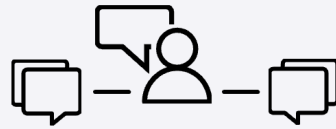
Amazon Bedrock Agents

Enable generative AI applications to execute multistep tasks using company systems and data sources



1

Select your
foundation model



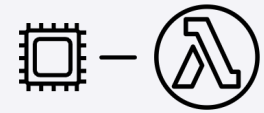
2

Provide basic
instructions



3

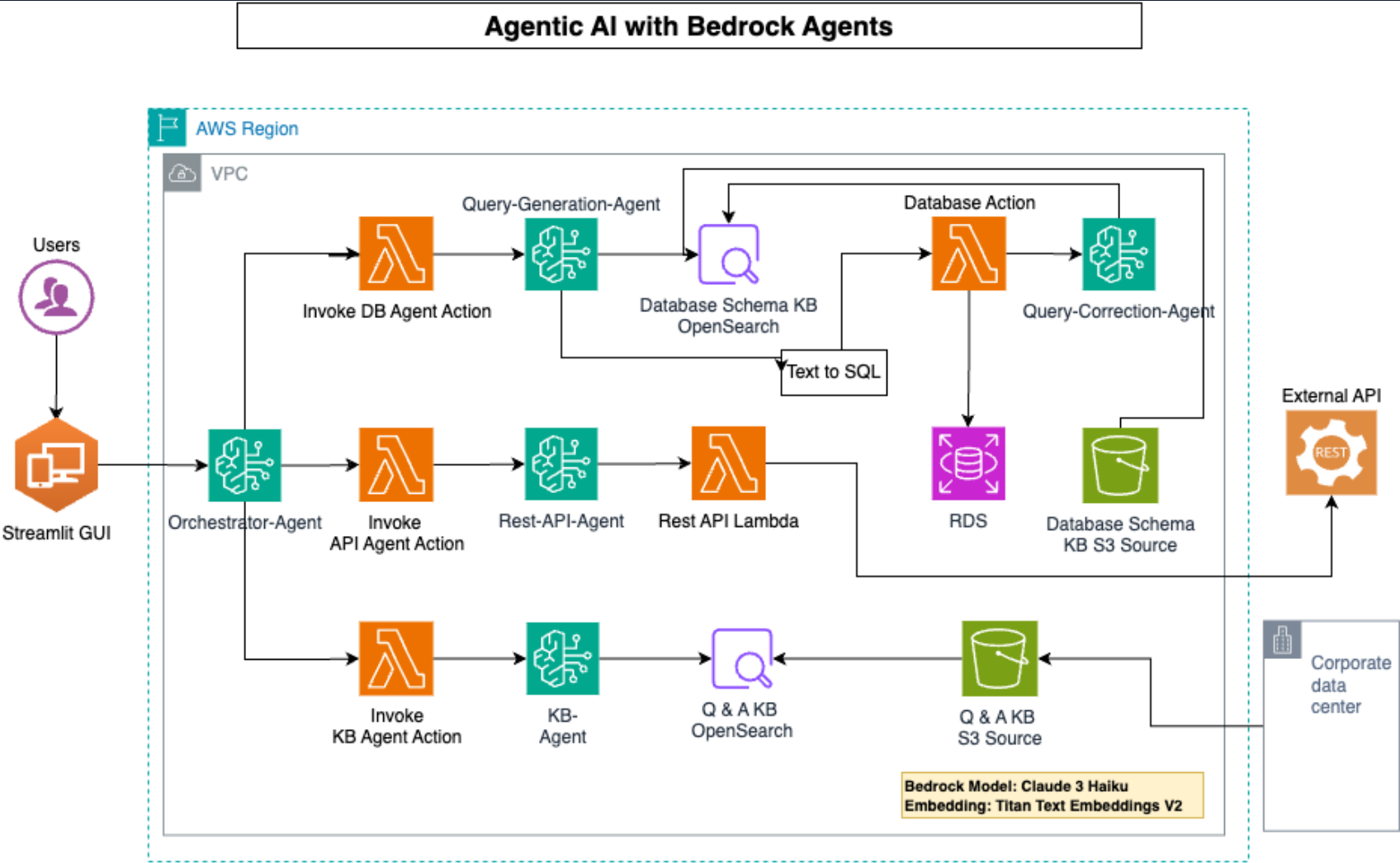
Select relevant
data sources



4

Specify available
actions

Serverless in Agentic AI



AWS Serverless MCP Server for AI-assistants

An [open-source tool](#) that combines AI assistance with serverless expertise to streamline how developers build serverless applications

Key benefits of the Serverless MCP Server include:

- **AI-powered serverless development:** Provides rich contextual information to AI coding assistants to ensure your serverless application aligns with AWS best practices.
- **Comprehensive tooling:** Offers tools for initialization, deployment, monitoring, and troubleshooting of serverless applications.
- **Architecture guidance:** Helps evaluate design choices and select optimal serverless patterns based on application needs. Offers recommendations on event sources, function boundaries, and service integrations.
- **Operational best practices:** Ensures alignment with AWS architectural principles. Suggests effective use of AWS services for event processing, data persistence, and service communication, and guides implementation of security controls, performance tuning, and cost optimization.
- **Security-first approach:** Implements built-in guardrails with read-only defaults and controlled access to sensitive data.

Takeaways

- **No infrastructure management** - AWS handles provisioning, scaling, and operations so developers focus on code
- **Event-driven architecture wins** - Asynchronous, loosely coupled systems provide better resilience and scalability than synchronous APIs
- **AWS leads serverless market** - Comprehensive service portfolio with 1M+ Lambda customers and 2T+ monthly EventBridge invocations
- **Multiple patterns for different needs** - API Gateway+Lambda for sync responses, EventBridge+Lambda for async processing, Step Functions for workflows
- **AI-integrated development** - Bedrock Agents for multi-step AI tasks, MCP Server for AI-assisted serverless development

Thank you!

Thanh Dang

Solution Architect, AWS Vietnam

www.linkedin.com/in/thanhmasan

<https://aws.amazon.com/serverless/>

